BIO201 CELL BIOLOGY MIDTERM MEGA FILE

ALL DATA FROM PAST PAPERS (2016-2022)

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1. **What is Kinetic energy?** 2

The energy possessed by an object because of its motion, equal to one half the mass of the body times the square of its velocity.

2. **Define Klinfelter syndrome?** 2

Klinefelter syndrome is a genetic condition affecting males, and it often isn't diagnosed until adulthood. Klinefelter syndrome may adversely affect testicular growth, resulting in smaller than normal testicles, which can lead to lower production of testosterone.

3. **Epistasis, co-dominance, incomplete dominance?** 5

Epistasis

Epistasis is a circumstance where the expression of one gene is modified by the expression of one or more other genes.

Co-dominance

Codominance, as it relates to genetics, refers to a type of inheritance in which two versions (alleles) of the same gene are expressed separately to yield different traits in an individual.

Incomplete dominance

Incomplete dominance results from a cross in which each parental contribution is genetically unique and gives rise to progeny whose phenotype is intermediate. Incomplete dominance is also referred to as semi-dominance and partial dominance.

4. **Define Mutation, Genetic engineering and plasmid?** 3

Mutation

A mutation is a change in the DNA sequence of an organism. Mutations can result from errors in DNA replication during cell division, exposure to mutagens or a viral infection.

Genetic Engineering

Genetic engineering (also called genetic modification) is a process that uses laboratory-based technologies to alter the DNA makeup of an organism. This may involve changing a single base pair (A-T or C-G), deleting a region of DNA or adding a new segment of DNA.

Plasmid

A plasmid is a small circular DNA molecule found in bacteria and some other microscopic organisms. Plasmids are physically separate from chromosomal DNA and replicate independently.

5. Difference between Klinfilter Syndrome and turner syndrome?

Klinefelter syndrome

Klinefelter syndrome is a genetic condition affecting males, and it often isn't diagnosed until adulthood. Klinefelter syndrome may adversely affect testicular growth, resulting in smaller than normal testicles, which can lead to lower production of testosterone.

Turner syndrome

Turner syndrome is a condition that affects only females, results when one of the X chromosomes (sex chromosomes) is missing or partially missing. Turner syndrome can cause a variety of medical and developmental problems, including short height, failure of the ovaries to develop and heart defects.

6. What is a lysosome?

Lysosomes are membrane-enclosed organelles that contain an array of enzymes capable of breaking down all types of biological polymers proteins, nucleic acids, carbohydrates, and lipids.

7. Main components of PCR? 3

The various components required for PCR include a DNA sample, DNA primers, free nucleotides called ddNTPs, and DNA polymerase.

8. Define coenzyme?

A coenzyme is defined as an organic molecule that binds to the active sites of certain enzymes to assist in the catalysis of a reaction.

9. Role of protein in living organisms?

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Proteins provide many of the structural elements of a cell, and they help to bind cells together into tissues. Proteins, in the form of antibodies, protect animals from disease, and many hormones are proteins. Proteins control the activity of genes and regulate gene expression.

10. Define Cross pollination? 2

Cross pollination is a natural method in which transfer of pollen takes place from an anther of a flower of one plant to a stigma of a flower of another plant of the same species.

11. Quaternary structure of protein? 2

Protein quaternary structure is the fourth classification level of protein structure. Protein quaternary structure refers to the structure of proteins which are themselves composed of two or more smaller protein chains.

12. Prosthetic group and give examples? 3

Prosthetic group is a type of a helper molecule which is a non-proteinaceous compound that helps enzymes to perform their functions.

For Example :

The heme group in hemoglobin is a prosthetic group.

13. Gel electrophoresis? 3

Gel electrophoresis is a laboratory method used to separate mixtures of DNA, RNA, or proteins according to molecular size. In gel electrophoresis, the molecules to be separated are pushed by an electrical field through a gel that contains small pores.

14. Active site? (2)

The active site is the region of an enzyme where substrate molecules bind and undergo a chemical reaction.

15. Sex linked disorder? (2)

A sex-linked inherited disorder is caused by a genetic mutation on the X chromosome. These are also inherited in a dominant or recessive pattern. There are also disorders associated with the Y chromosome, but these are inherited less frequently. That is because they are often associated with male infertility.

16. Steps of mitosis? (3)

Mitosis is that step in the cell cycle where the newly formed DNA is separated and two new cells are formed with the same number and kind of chromosomes as the parent nucleus.

Mitosis is involving five phases, based on the physical state of the chromosomes and spindle.

i. Prophase

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ii. Pro-metaphase
iii. Metaphase
iv. Anaphase
v. Telophase.

17. PKU means and its caused? (3)

Phenylketonuria also called PKU, is a rare inherited disorder that causes an amino acid called phenylalanine to build up in the body. PKU is caused by a change in the phenylalanine hydroxylase (PAH) gene. This gene helps create the enzyme needed to break down phenylalanine.

18. Requirements of PCR? (5)

- **DNA Template:**

The DNA of interest from the sample.

- **DNA Polymerase:**

Taq Polymerase is used. It is thermostable and does not denature at very high temperatures.

- **Oligonucleotide Primers:**

These are the short stretches of single-stranded DNA complementary to the 3' ends of sense and anti-sense strands.

- **Deoxyribonucleotide triphosphate:**

These provide energy for polymerization and are the building blocks for the synthesis of DNA. These are single units of bases.

- **Buffer System:**

Magnesium and Potassium provide optimum conditions for DNA denaturation and renaturation. It is also important for fidelity, polymerase activity, and stability.

19. Define Cell division? (2)

Cell division is the process by which a parent cell divides, when a mother cell divides into two or more daughter cells. Cell division usually occurs as part of a larger cell cycle.

20. Define nucleoside and nucleotide? (3)

Nucleoside

A nucleoside is a compound formed by the union of a nitrogen base with a pentose sugar.

Nucleotide

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A nucleotide is a compound formed by the union of a nitrogen base, a nitrogen sugar and phosphate.

21. Features of Mitochondria?

Mitochondria are typically round to oval in shape and range in size from 0.5 to 10 μm. In addition to producing energy, mitochondria store calcium for cell signalling activities, generate heat, and mediate cell growth and death.

Further features of Mitochondria involve:

- Production of ATP
- Calcium Homeostasis
- Regulation of Innate Immunity
- Programmed Cell Death
- Stem Cell Regulation

22. What is DNA Fingerprinting?

DNA fingerprinting is a method used to identify an individual from a sample of DNA by looking at unique patterns in their DNA.

23. Functions of DNA?

The function of DNA is to store all of the genetic information that an organism needs to develop, function, and reproduce. DNA is involved in:

- Replication process
- Mutations
- Transcription
- Cellular Metabolism
- DNA Fingerprinting
- Gene Therapy

24. What is smooth endoplasmic reticulum?

SER is a membrane-bound network of tubules without surface ribosomes. SER is not involved in protein synthesis. Its main function is the synthesis of lipids, steroids, and carbohydrates, as well as the metabolism of exogenous substances, such as drugs or toxins.

25. Difference between purines and pyrimidines?

Purines

- The purines in DNA are adenine and guanine, the same as in RNA.
- Purines are larger than pyrimidines because they have a two-ring structure.

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Pyrimidines

- The pyrimidines in DNA are cytosine and thymine in RNA, they are cytosine and uracil.
- Pyrimidines only have a single ring.

26. Function of cellular membrane?

- Protecting the cell against physical damage and invading pathogens.
- Cell wall controls and regulates the direction of cell growth.
- Providing the strength, structural support and maintaining the shape of the cell.
- Functions as a storage unit by storing carbohydrates for use in plant growth, especially in seeds.
- It allows entry of smaller molecules through it freely.

27. Components of Cellular Membrane? 5

The principal components of the cell membrane are:

- Lipids
- Phospholipids
- Cholesterol
- Proteins
- carbohydrate groups that are attached to some of the lipids and proteins.

28. PCR techniques? 5 marks

PCR or Polymerase Chain Reaction is a technique used in molecular biology to create several copies of a certain DNA segment. This technique was developed in 1983 by Kary Mullis, an American biochemist. PCR has made it possible to generate millions of copies of a small segment of DNA. This tool is commonly used in the molecular biology and biotechnology labs. The PCR technique is based on the enzymatic replication of DNA. In PCR, a short segment of DNA is amplified using primer mediated enzymes. DNA Polymerase synthesises new strands of DNA complementary to the template DNA. The DNA polymerase can add a nucleotide to the pre-existing 3'-OH group only. Therefore, a primer is required. Thus, more nucleotides are added to the 3' prime end of the DNA polymerase.

29. Define semi conservative replication, Conservative replication and dispersive replication of DNA? (5)

- **Semiconservative replication**

Semiconservative replication describes the mechanism of DNA replication in all known cells. DNA replication occurs on multiple origins of replication along the DNA template strand. As the DNA double helix is unwound by helicase, replication occurs separately on each template strand in antiparallel directions.

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- **Conservative Replication**

In conservative replication, two DNA copies are produced from one original DNA, which serves as a template. Out of these two, one is entirely new DNA, and the other is made of old DNA strands.

- **Dispersive replication**

It is a replication process that also produces two DNA copies from the original DNA. Here, the two DNA products have distinct regions composed of new and original strands. This replication happens in a random fashion.

30. Define Ester Linkage?

The bond formed between both organic molecules is called an ester linkage.

OR

In an ester molecule, the bond connecting the atom doubly bonded to oxygen and the oxygen atom bearing the alkyl or aryl group is called the ester bond or, in biochemistry, ester linkage.

31. Function of PAH enzyme?

The PAH gene provides instructions for making an enzyme called phenylalanine hydroxylase. This enzyme is responsible for the first step in processing phenylalanine, which is a building block of proteins an amino acid obtained through the diet.

32. What are X-linked disorders?

All sex-linked disorders are X-linked. There are very few X-linked dominant diseases, inheritance pattern is characterized by transmission of the disease to 50% of the sons and daughters of an affected female. For example: Bruton's tyrosine kinase, a protein whose mutation is associated with X-linked agammaglobulinemia.

33. Proteins denaturation?

Since **denaturation** reactions are not strong enough to break the peptide bonds, the primary structure (sequence of amino acids) remains the same after a denaturation process. **Denaturation** disrupts the normal alpha-helix and beta sheets in a **protein** and uncoils it into a random shape.

34. Down syndrome define?

Down syndrome (DS or DNS), also known as trisomy 21, is a genetic disorder caused by the presence of all or part of a third copy of chromosome 21. It is typically associated with physical growth delays, characteristic facial features, and mild to moderate intellectual disability.

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35. Describe the phases of mitosis?

There are essentially six stages of mitosis (some textbooks tend to club pro-metaphase and prophase into a single stage). The stages of mitosis comprise:

1. Interphase
2. Prophase
3. Pro-metaphase
4. Metaphase
5. Anaphase
6. Telophase

Interphase

Technically, the interphase is not a part of mitosis, however, it is still a crucial process as it leads up to the process of mitosis. Hence, the interphase refers to all the other stages of cell cycle other than mitosis.

Prophase

Prophase is technically the first stage of mitosis. It is also the longest phase of mitosis, where the chromatin condenses into chromosomes and the nuclear membrane breaks down (in **pro-metaphase**). In animal cells, the centrioles located near the nucleus begin to split and move to the opposite ends of the cell.

Metaphase

In metaphase, the microtubules pull on the chromosomes with equal force, and the chromosome moves to the center of the cell.

Anaphase

The anaphase is marked by the splitting of the sister chromatids. These sister chromatids then become the chromosomes of the daughter nuclei.

Telophase

The telophase is the final phase of mitosis. It begins after the replicated, paired chromosomes are separated and pulled at the opposite ends of the pole.

36. Translation in detail?

Translation is the process by which the genetic code contained within a messenger RNA (mRNA) molecule is decoded to produce a specific sequence of amino acids in a polypeptide chain.

It occurs in the **cytoplasm** following DNA transcription and, like transcription, has three stages: initiation, elongation and termination. In this article we will discuss the components and stages of DNA translation.

Components of Translation

The key components required for translation are mRNA, ribosomes, and transfer RNA (tRNA).

During translation, mRNA nucleotide bases are read as **codons** of three bases. Each codon codes for a particular amino acid. Every tRNA molecule possesses an **anticodon** that is complementary to the mRNA codon, and at the opposite end lies the attached amino acid. tRNA molecules are therefore responsible for bringing amino acids to the ribosome in the correct order, ready for polypeptide assembly.

A single amino acid may be coded for by more than one codon. There are also specific codons that signal the start and the end of translation.

Aminoacyl-tRNA synthetases are enzymes that link amino acids to their corresponding tRNA molecules. The resulting complex is charged and is referred to as an aminoacyl-tRNA.

Steps of translation: -

- Initiation
- Elongation
- Termination

37.note on transcription (5)

Transcription is the process by which the information in a strand of DNA is copied into a new molecule of messenger RNA (mRNA). DNA safely and stably stores genetic material in the nuclei of cells as a reference, or template. Meanwhile, mRNA is comparable to a copy from a reference book because it carries the same information as DNA but is not used for long-term storage and can freely exit the nucleus. Although the mRNA contains the same information, it is not an identical copy of the DNA segment, because its sequence is complementary to the DNA template.

Transcription is carried out by an enzyme called RNA polymerase and a number of accessory proteins called transcription factors. Transcription factors can bind to specific DNA sequences called enhancer and promoter sequences in order to recruit RNA polymerase to an appropriate transcription site. Together, the transcription factors and RNA polymerase form a complex called the transcription initiation complex. This complex initiates transcription, and the RNA polymerase begins mRNA synthesis by matching complementary bases to the original DNA

strand. The mRNA molecule is elongated and, once the strand is completely synthesized, transcription is terminated. The newly formed mRNA copies of the gene then serve as blueprints for protein synthesis during the process of translation.

38.Techniques of genetic engineering (long)

Genetic Engineering

Genetic engineering (also called genetic modification) is a process that uses laboratory-based technologies to alter the DNA makeup of an organism. This may involve changing a single base pair (A-T or C-G), deleting a region of DNA or adding a new segment of DNA.

Techniques of Genetic Engineering: -

i. Process
ii. History
iii. Choosing target genes
iv. Gene manipulation
v. Extraction from cells
vi. Gene isolation
vii. Modification
viii. Transformation
ix. Development
x. Selection
xi. Regeneration
xii. Confirmation
xiii. Gene targeting
xiv. Gene trapping

39. Cell cycle (long)

The cell cycle is the process a cell undertakes to replicate all of its genetic material and divide into two identical cells. In this article, we will look at the different stages of the cell cycle and what happens in each stage. We will also consider the regulation of the cell cycle, and look at some examples of its dysregulation.

Phases of the Cell Cycle

The cell cycle is a 4-stage process consisting of Gap 1 (G1), synthesis (S), Gap 2 (G2) and mitosis (M), which a cell undergoes as it grows and divides. After completing the cycle, the cell either starts the process again from G1 or exits the cycle through G0. From G0, the cell can undergo terminal differentiation.

The stages in the cell cycle between one mitosis and the next, which include G1, S and G2, are known collectively as the **interphase**.

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1. **G1 phase: -**
   - Cell increases in size
   - Cellular contents are duplicated

2. **S phase: -**
   - DNA replication
   - Each of the 46 chromosomes are replicated

3. **G2 phase: -**
   - Cell grows more
   - Organelles are proteins are developing in preparation for cell division

4. **M phase: -**
   - Mitosis followed by cytokinesis
   - Formation of two identical sister chromatids

5. **G◦ Phase: -**
   While some cells are constantly dividing, some cell types are **quiescent**. These cells exit G1 and enter a resting state called G0. In G0, a cell is performing its function without actively preparing to divide. G0 is a permanent state for some cells, while others may re-start division if they get the right signals

40. 3 dilemmas of DNA replication

The three steps in the process of DNA replication are initiation, elongation and **termination.** Replication depends on the pairing of bases between the two strands of DNA.

- Initiation.
- Elongation.
- Termination.

41.mitochondrial features

Mitochondria is double membrane-enclosed organelle. lt contains its own DNA,ribosomes, and other components required for protein synthesis, most of its proteins are encoded in the cell nucleus and imported from the cytosol There are two sub-compartments in mitochondria: the internal matrix space and the intermembrane space.

42.restriction endonucleases

Special bacterial enzyme can cut double stranded DNA at specific sites in a test tube. Restriction endonucleases (restriction enzymes) act as a kind of immune system, protecting the cell from the invasion of foreign DNA (virus). Many recombinant DNA technologies, which the field of

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biotechnology heavily relies on, are unlikely to have been developed without the discovery of restriction enzymes

43.macromolecules present in cell.

The four major types of **macromolecules found** in living cells—carbohydrates, lipids, proteins, and nucleic acids--are made of these smaller, repeating subunits called monomers. The monomers within one molecule are not always identical but they always have similar chemical structures.

44. Chargaff's Rule?

Chargaff's rules state that DNA from any cell of all organisms should have a 1:1 ratio (base Pair Rule) of pyrimidine and purine bases and, more specifically, that the amount of guanine should be equal to cytosine and the amount of adenine should be equal to thymine.

45.What type of bacterial colonies F. Griffith was used in his experiment? (2 marks)

Griffith used two strains of pneumococcus (Streptococcus pneumoniae) bacteria which infect mice — a type III-S (smooth) which was virulent, and a type II-R (rough) strain which was nonvirulent. The III-S strain synthesized a polysaccharide capsule that protected itself from the host's immune system, resulting in the death of the host, while the II-R strain did not have that protective capsule and was defeated by the host's immune system.

46.difference between DNA and RNA? (5)

Structurally, DNA and RNA are nearly identical. As mentioned earlier, however, there are three fundamental **differences** that account for the very different functions of the two molecules. RNA has a ribose sugar instead of a deoxyribose sugar like DNA.RNA nucleotides have a uracil base instead of thymine.

47.Describe the techniques used by Mendel in his experiment (2)

Gregor Mendel conducted hybridization experiments on around 29,000 pea plants. Peas were an ideal choice for Mendel to use because they had easily observable traits there were 7 of which he could manipulate. He began his **experiments** on peas with two conditions.

48.define aqua porins

Aquaporin: A water channel.

Aquaporins form pores in the membranes of cells and selectively conduct water molecules through the membrane, while preventing the passage of ions (such as sodium and potassium) and other small molecules. Aquaporins are typically composed of identical subunit proteins.

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49.law of inheritance

Gregor Mendel, through his work on pea plants, discovered the fundamental **laws of inheritance.** He deduced that genes come in pairs and are **inherited as** distinct units, one from each parent. Mendel tracked the segregation of parental genes and their appearance in the offspring as dominant or recessive traits.

50.role of lipids

The main biological **functions of lipids** include storing energy, signaling, and acting as structural components of cell membranes. ... Although the term "lipid" is sometimes used as a synonym for fats, fats are a subgroup of **lipids** called triglycerides.

51.stages of meiosis 1

the **stages of meiosis I are:**

- prophase I
- metaphase I
- anaphase I
- telophase I

52.ENLIST CELL organelles

Nucleolus.
Nucleus.
Ribosome (little dots)
Vesicle.
Rough endoplasmic reticulum. Golgi apparatus (or "Golgi body")
Cytoskeleton.
Smooth endoplasmic reticulum.

53.define threshold effect and write the rate of occurrence first degree of relative and identical twins?

A threshold effect is a sudden and radical change in a phenomenon which often occurs after surpassing a quantitative limit, called the **threshold.**
It may refer to:
Re-normaIization group, Threshold effect, a particle physics calculation. **Threshold effect** (genetics), a trait in genetics. Twins are two offspring produced by the same pregnancy. Twins

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can be either monozygotic **("identical"),** meaning that they develop from one zygote, which splits and forms two embryos, or dizygotic ("fraternal"), meaning that they develop from two different eggs.

54.Abbreviation of VNTRs

Variable Number of Tandem Repeats (genetics)

55.Ionic bond

Ionic bonding is a type of chemical bonding that involves the electrostatic attraction between oppositely charged ions, and is the primary interaction occurring in ionic compounds

56.Define carbohydrates. Also write their classification.

Carbohydrates: -

a large group of organic compounds occurring in foods and living tissues and including sugars, starch, and cellulose. They contain hydrogen and oxygen in the same ratio as water (2:1) and typically can be broken down to release energy in the animal body.

Classification: -

- Monosaccharides
- Disaccharides
- Oligosaccharides
- Polysaccharides

57. Define polyploidy and aneuploidy with examples.

Polyploidy: -
Cells (and their owners) are **polyploid** if they contain more than two haploid (n) sets of chromosomes; that is, their chromosome number is some multiple of n greater than the 2n content of diploid cells. For example, triploid (3n) and tetraploid cell (4n) cells are **polyploid.**

Chromosomes in Down syndrome, the most common human condition due to **aneuploidy.** Notice the three copies of chromosome 21 in the last row.

Aneuploidy: -
Aneuploidy is the presence of an abnormal number of chromosomes in a cell, for **example** a human cell having 45 or 47 chromosomes instead of the usual 46.

58.function of cell (2)

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The human body is composed of trillions of cells. They provide **structure** for the body, take in nutrients from food, convert those nutrients into **energy,** and carry out specialized functions. Cells also contain the body's hereditary material and can make copies of themselves.

59. types of cristae Golgi apparatus (2)

- Cristae Membrane
- Cristae Junctions

60. lysosome and heat shock protein

Lysosome: an organelle in the cytoplasm of eukaryotic cells containing degradative enzymes enclosed in a membrane.

Heat Shock Protein: a protein induced in a living cell in response to a rise in temperature above the normal level.

61. What are Helicase and Primase?

A **helicase—primase** complex (also **helicase-primase,** Hel/Prim, H-P or H/P) is a complex of enzymes including DNA **helicase** and DNA **primase.** The complex is used by herpesviruses in which it is responsible for lytic DNA virus replication

62. Electro-negativity

Electronegativity The tendency of an atom to attract electrons towards itself.

63. Difference b/w meiosis and mitosis?

The processes **differ** in two fundamental. **Meiosis** has two rounds of genetic separation and cellular division while **mitosis** only has one of each. In meiosis homologous chromosomes separate leading to daughter cells that are not genetically identical two cells with no net change in the number of chromosomes.

64. UV light sterilization

UV light sterilization is an environmentally friendly method of killing bacteria, mold, fungi, and viruses without the use of harmful chemicals and does not produce corrosive materials or disinfection by-products(DBPs).

65. Differentiate between T-cell and B-cell

An important difference between T-cells and B-cells is that B-cells can connect to antigens right on the surface of the invading virus or bacteria. This is different from T-cells, which can only connect to virus antigens on the outside of infected cells.

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Difference between T cell epitope and B cell epitope

	T cell epitope	B cell epitope
Receptor	TCR	BCR
Nature	short peptide	proteins, polysaccharides
Size	8-17 amino acid residues	5-15 amino acid residues or 5-7 monosaccharides
Types	linear epitope	conformational epitope or linear epitope
Position	any position in antigen	mostly exist on the surface of antigen

66. Differentiate between homotypic and heterotypic fusion

Homotypic cell fusion

Homotypic cell fusion occurs between cells of the same type. An example of this would be osteoclasts or myofibers being fusing together with their respective type of cells. Whenever the two nuclei merge a synkaryon is produced. Cell fusion normally occurs with nuclear fusion, but in the absence of nuclear fusion, the cell would be described as a bi-nucleated heterokaryon. A heterokaryon is the melding of two or more cells into one and it may reproduce itself for several generations. If two of the same type of cells fuse, but their nuclei do not fuse, then the resulting cell is called asyncytium.

Heterotypic cell fusion

Heterotypic cell fusion occurs between cells of different types, making it the exact opposite of homotypic cell fusion. There result of this fusion is also a synkaryon produced by the merging of the nuclei, and a bi-nucleated heterokaryon in the absence of nuclear fusion. An example of this would be BMDCs being fused with parenchymatous organs.

67. Types of antibiotics

These are the main types of antibiotics.

Penicillins: such as penicillin and amoxicillin

Cephalosporins: such as cephalexin (Keflex)

Macrolides: such as erythromycin (EMycin), clarithromycin (Biaxin), and azithromycin (Zithromax)

Fluoroquinolones: such as ciprofloxacin (Cipro), levofloxacin (Levaquin), and ofloxacin (Floxin)

Sulfonamides: such as co-trimoxazole (Bactrim) and trimethoprim (Proloprim)

Tetracyclines: such as tetracycline (Sumycin, Panmycin) and doxycycline (Vibramycin)

Aminoglycosides: such as gentamicin (Garamycin) and tobramycin (Tobrex)

68. Confluent monolayer

Confluent monolayer refers to cells in tissue culture, e.g. epithelial cells, which form a cohesive sheet comprising of a single cell layer filling the entire surface area of the bottom of the culture dish. Cells have then reached confluence.

69. Types of RNA and their functions (3)

- **mRNA**: mRNA is transcribed from DNA and contains the genetic blueprint to make proteins
- **tRNA**: tRNAs are RNA molecules that translate mRNA into proteins.
- **Rrna:** rRNA forms ribosomes, which are essential in protein synthesis. A ribosome contains a large and small ribosomal subunit.

70. Difference between PKU and PAH (2)

PKU: Phenylketonuria (PKU) is an autosomal recessive genetic disorder characterized by a deficiency in the enzyme phenylalanine hydroxylase (PAH)

PAH: is the enzyme which causes PKU

71. Note on endocytosis, pinocytosis & phagocytosis (5)

- **Endocytosis: -**

The material to be ingested is progressively enclosed by a small portion of the plasma membrane, which first invigilates and then pinches off to form an endocytic vesicle containing the ingested substance or particle.

- **Pinocytosis: -**

Pinocytosis involves ingestion of fluid & solutes via small pinocytic vesicles (about 100 nm in diameter)

- **Phagocytosis: -**

Phagocytosis involves the ingestion of large particles microorganisms or dead cells via large vesicles called phagosomes (>250 nm in diameter)

72. What is Alpaktonuria?

Alkaptonuria, or black urine disease, is a very rare inherited disorder that prevents the body fully breaking down two protein building blocks (amino acids) called tyrosine and phenylalanine. It results in a build-up of a chemical called homogentisic acid in the body.

73. What is Codon?

Codon is a sequence of three nucleotides which together form a unit of genetic code in a DNA or RNA molecule.

74. Write three stop codons?

- UAG
- UAA
- UGA

75. Write three RNA Viruses?

- Orthomyxoviruses
- Hepatitis C Virus (HCV)
- Ebola disease
- SARS
- influenza
- polio measles
- retrovirus

76. sites of larger subunits of Ribosome?

The large ribosomal subunit has three places that can bind tRNA:

- the A site
- the P site
- the E site.

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Separate Whatsapp Groups for Girls and Boys

Special Internship Group with Senior Guideline

Zero Percentage Irrelevant Chatting in study groups

Follow this Link for Joining Study Group:

https://chat.whatsapp.com/D2yLdI6LFFpJuS2EKWYmbE

Remember Us in Your Prayers

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